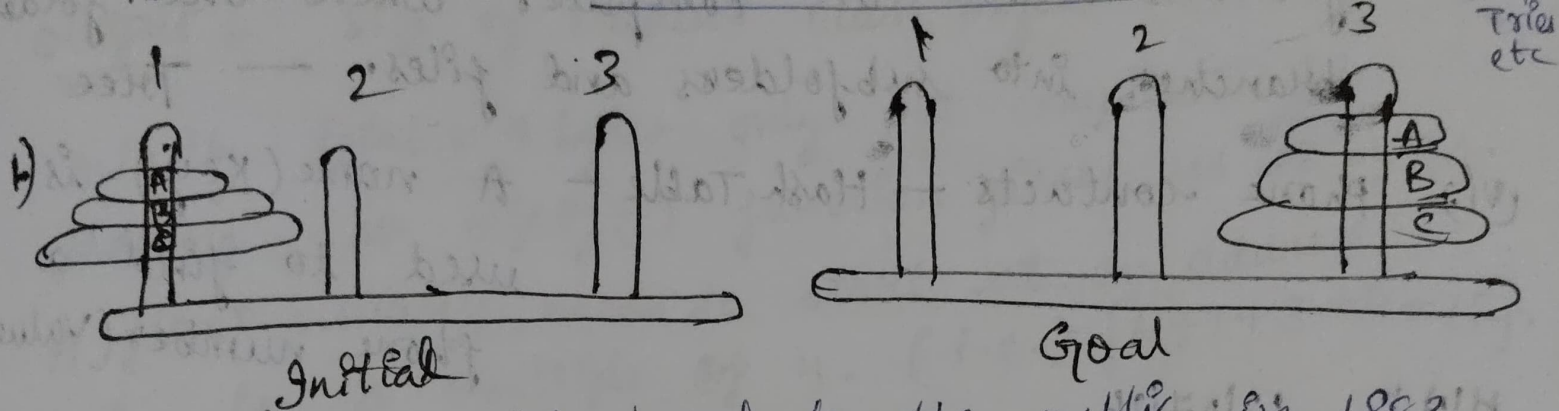


HW:- 10/07/2026

- 1.) Find out the history of Towers of Hanoi.
 - 2.) Today from the morning 7:30 AM to 4:30 pm.
List out various Data structures in your surroundings.
-



The puzzle was introduced to the public in 1883 by the French Mathematician Edouard Lucas. The myth: Lucas accompanied it with a romanticized legend about an Indian temple in Varanasi.

Inside the great temple of Brahma, beneath the dome that marks the centre of the ~~universe~~ world. They say During creation, God placed 64 golden disks of decreasing size onto one of the three diamond needles which sits a brass plate supporting those needles. So a group of priests work night and day, constantly moving the disks according to rules. You can move only one disk at a time. and larger disk cannot sit on top of smaller one. They mainly say that the moment when the priests complete the transfer of all 64 disks to the third needle, "the world will end".

They say minimum number of moves required to solve a Tower of Hanoi puzzle. with "n" disks is given by the formula :- $[2^n - 1]$

It is used in modern world to teach recursion.

- Q.)
- (i) stack of plates - Stack - Last in, first out
 - (ii) Queue at bus stop - Queue - First in, first out
 - (iii) Student marks list - Array - Each element can be accessed using its position (index).
 - (iv) Music playlist - Linked List - Elements are connected one after the other to make it easy for adding and removing items.
 - (v) Road maps or google maps - Graphs
 - (vi) folder structure on computer where one folder branches into subfolders and files. - Tree
 - (vii) phone contacts - Hash Table - A name (Key) is used to find a phone number (Value).

HW :- 12/07/26

1) Write a Cpp code which encloses a class & its member.